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Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2024

Code: 104606		***Time: 03 Hours***
Semester: VI	***Subject: Digital CMOS VLSI Design***	***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only) **[2 x 7 = 14]**

(a) Which is the first step in CMOS fabrication?
- (i) Lithography
- (ii) Oxidation
- (iii) Diffusion
- (iv) Ion implantation

(b) nWell process is commonly used in:
- (i) PMOS fabrication
- (ii) NMOS fabrication
- (iii) CMOS fabrication
- (iv) Bipolar fabrication

(c) In a MOS transistor, the current between drain and source flows due to the movement of:
- (i) Electrons only
- (ii) Holes only
- (iii) Electrons and holes
- (iv) Neither electrons nor holes

(d) What controls the conductivity in a MOSFET?
- (i) Source voltage
- (ii) Drain voltage
- (iii) Gate voltage
- (iv) Substrate

(e) Which of the following loads is used in a CMOS inverter?
- (i) Resistor
- (ii) Depletion-mode MOSFET
- (iii) Enhancement-mode MOSFET
- (iv) Capacitor

(f) Inverter delay time can be minimized by:
- (i) Increasing supply voltage
- (ii) Reducing load capacitance
- (iii) Both a and b
- (iv) None of the above

(g) Which of the following gates is a basic building block of CMOS logic?
- (i) AND gate
- (ii) NOT gate
- (iii) XOR gate
- (iv) NAND gate

(h) The logic level of a MOS circuit is determined by:
- (i) Gate voltage
- (ii) Source voltage
- (iii) Drain voltage
- (iv) Substrate voltage

(i) In MOS logic circuits, a pass gate is constructed using:
- (i) Only PMOS
- (ii) Only NMOS
- (iii) Both NMOS and PMOS
- (iv) None of the above

(j) What does MOS stand for in MOSFET?
- (i) Metal Oxide Semiconductor
- (ii) Metal Oxide Silicon
- (iii) Metal Organic Semiconductor
- (iv) Metal Oxide Structure

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Explain the structure and operation of a MOSFET with a neat diagram.

(b) How does the nWell process differ from the pWell process in CMOS technology?

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Describe the current-voltage characteristics of MOSFETs in different operating regions.

(b) Describe the CMOS fabrication process in detail, highlighting each step.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Compare the operation of n-channel and p-channel MOSFETs.

(b) Explain the importance of mask layers in CMOS fabrication and their design considerations.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) What is switching power dissipation in CMOS inverters? Derive the formula and explain the factors affecting it.

(b) Explain the working principle of a CMOS NAND gate with a circuit diagram.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Explain the voltage transfer characteristics of a CMOS inverter.

(b) Design a 4-input CMOS logic circuit for the AND-OR-INVERT function and explain its working.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the factors affect the delay time of a MOS inverter and how it can be optimized.

(b) Explain the role of clock signals in sequential logic circuits and its impact on circuit performance.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the advantages and disadvantages of MOS logic circuits having depletion-mode nMOS loads.

(b) Design a clocked CMOS D flip-flop and explain its operation.

Q.9 Solve both questions : **[7 x 2 = 14]**

(a) Explain the operation of a CMOS SR latch with a detailed circuit diagram.

(b) Describe how interconnect parasitics impact the performance of MOS inverters.

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